

EEG Analysis Report

I. Patient Data

Patient Name : None

Patient ID : None

Measurement Date : 2019-05-06 14:00:30.520000+00:00

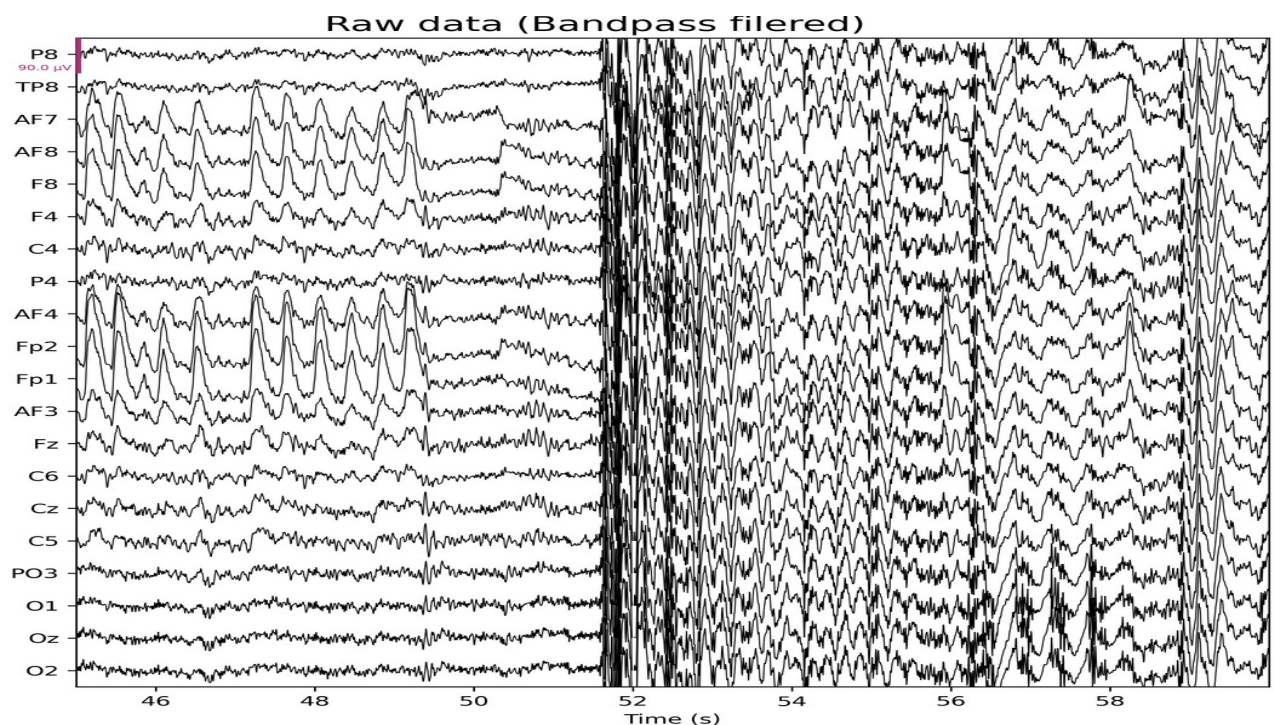
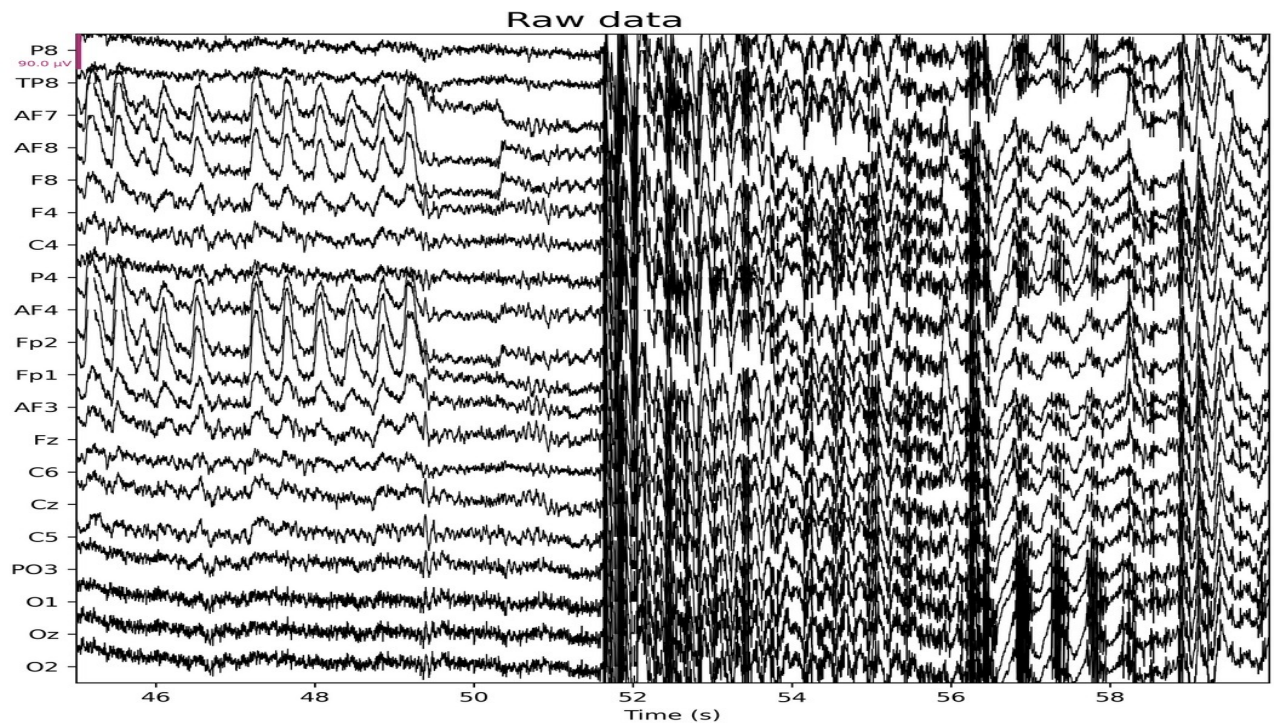
Sampling Frequency : 500.0 Hz

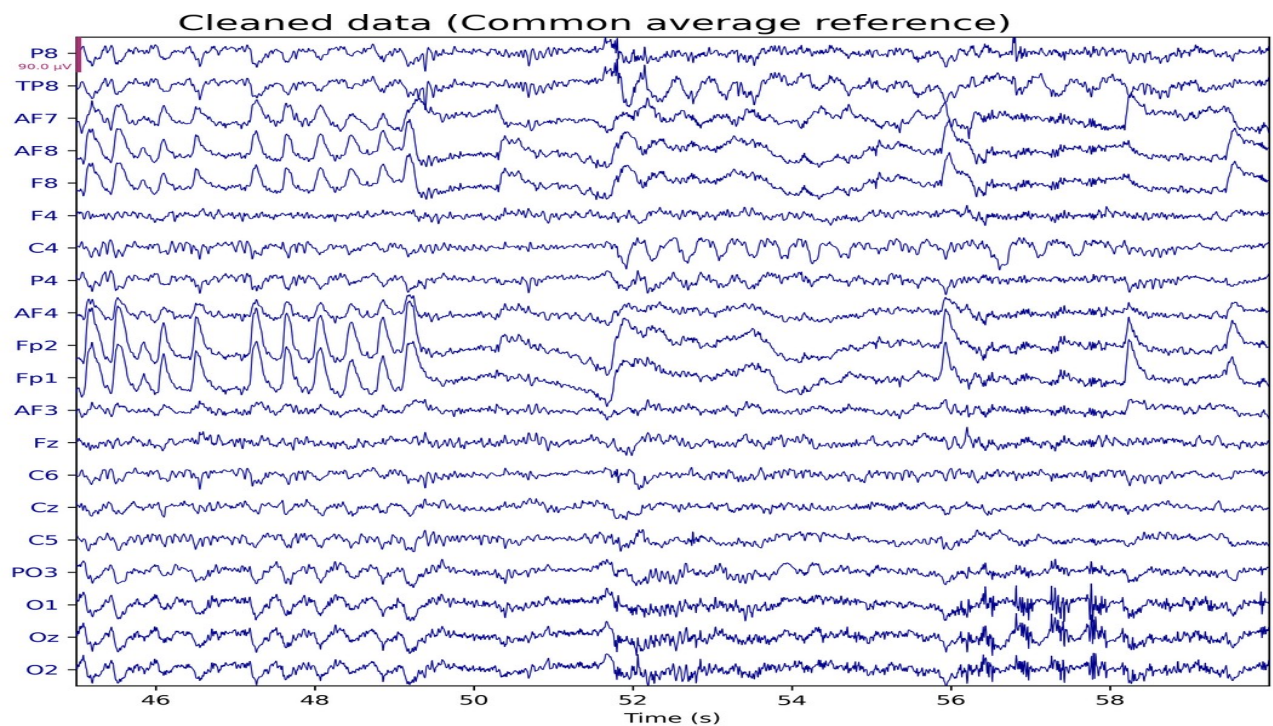
Number of Channels : 33

Channel Names : P8, TP8, AF7, AF8, F8, F4, C4, P4, AF4, Fp2, Fp1, AF3, Fz, C6, Cz, C5, PO3, O1, Oz, O2, PO4, Pz, Fpz, FC6, P3, C3, F3, F7, FC5, FCz, TP7, P7, STI 014

II. EEG

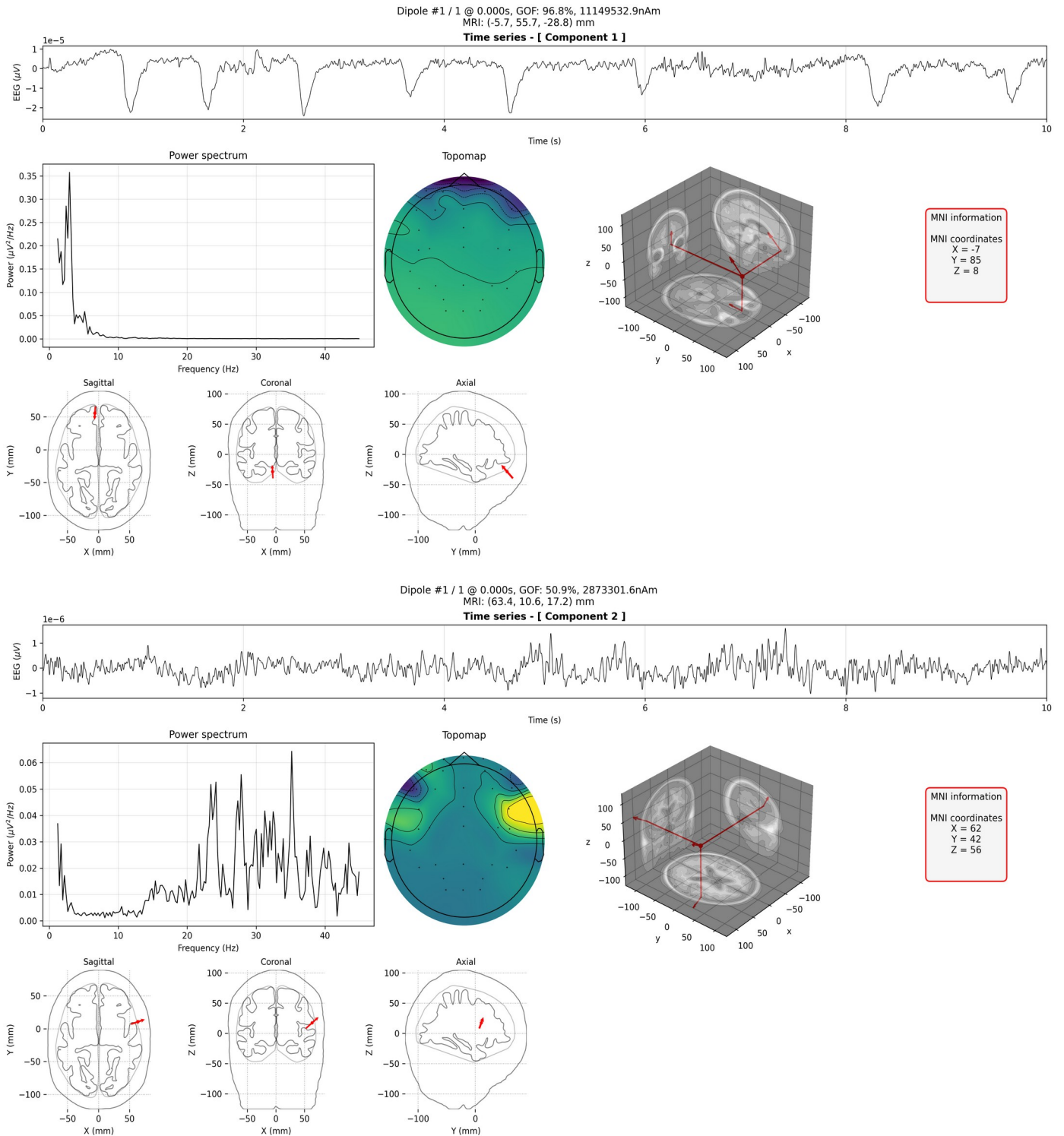
EEG data is a record of the oscillations of electrical brain potentials recorded from electrodes on the human scalp (T100) (T101) (T005). The raw data in the figure below have been cleaned by the application of high-pass and low-pass filters.

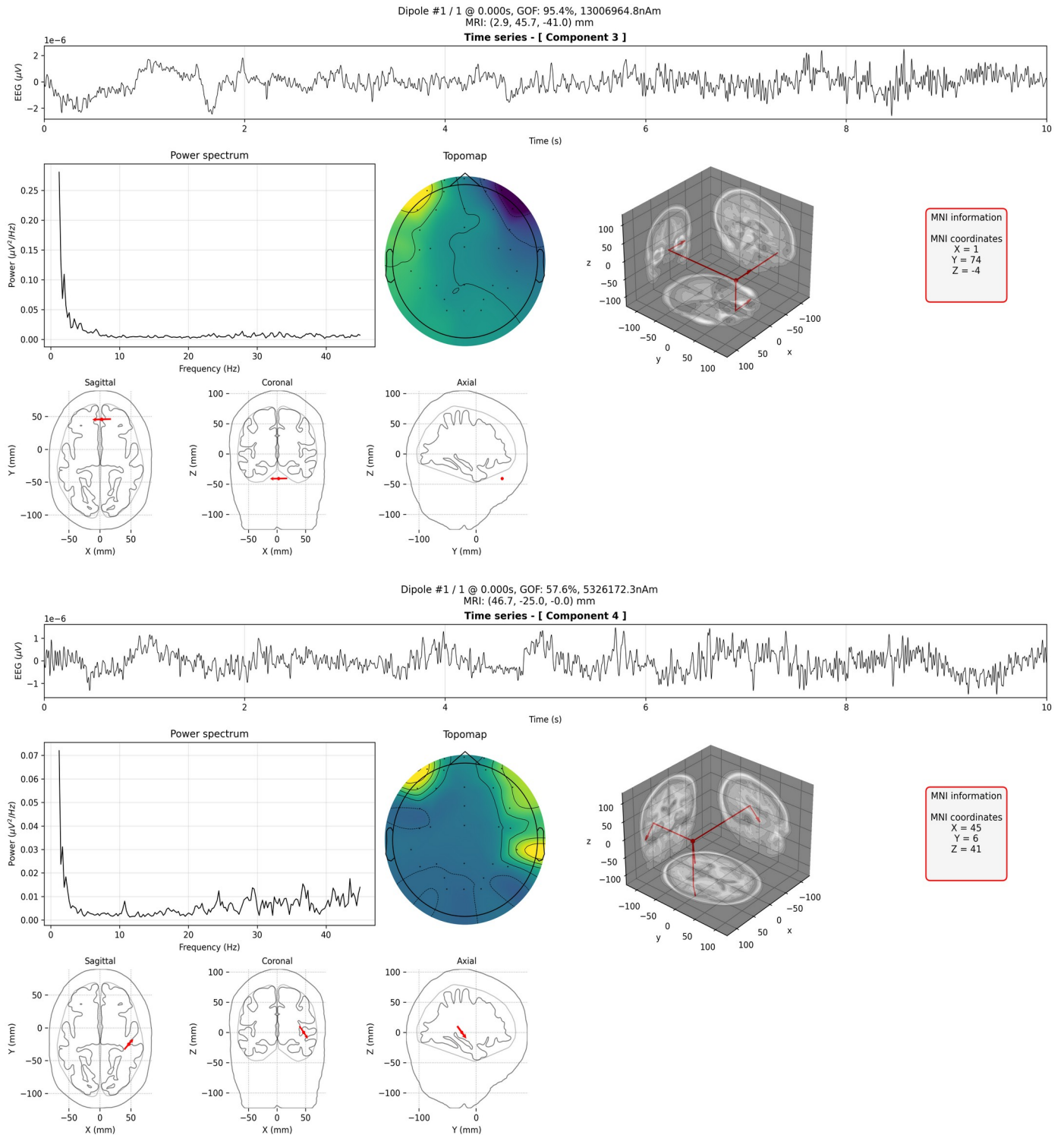


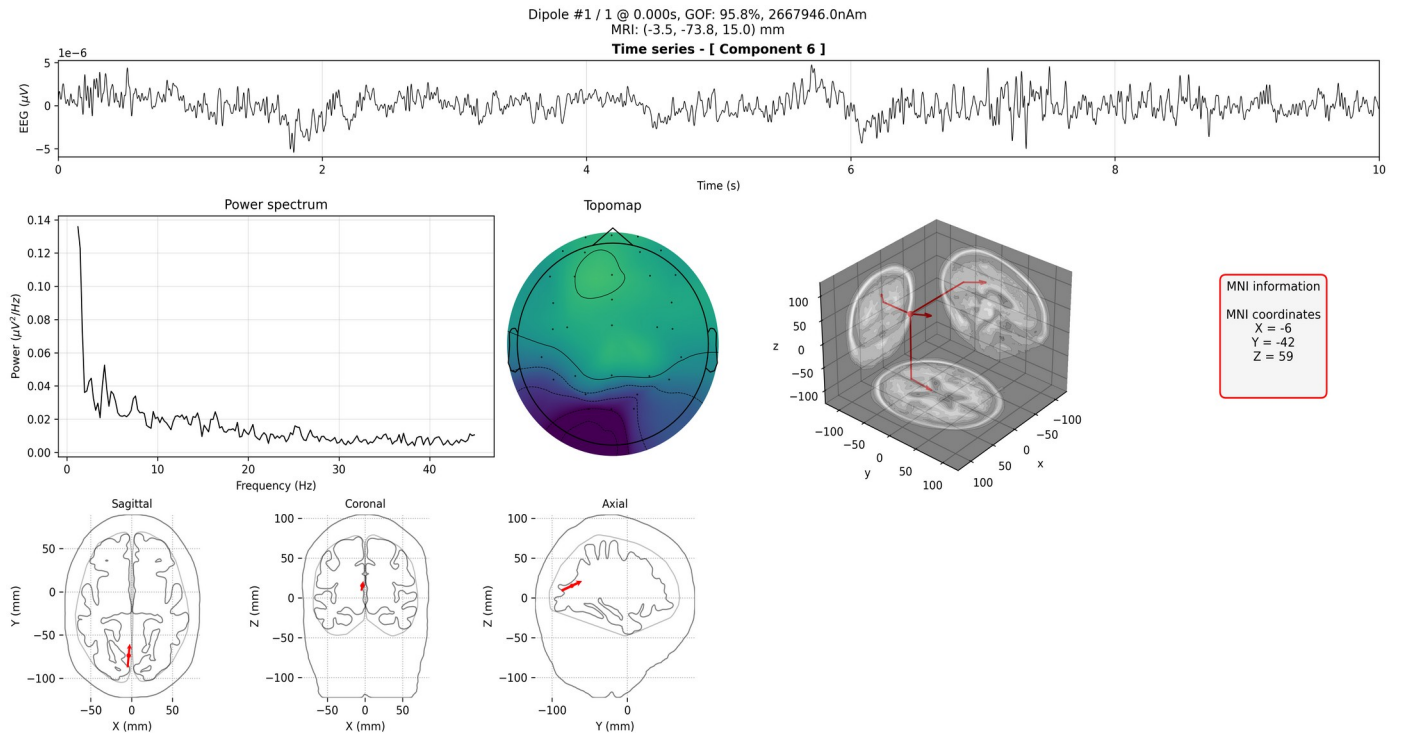
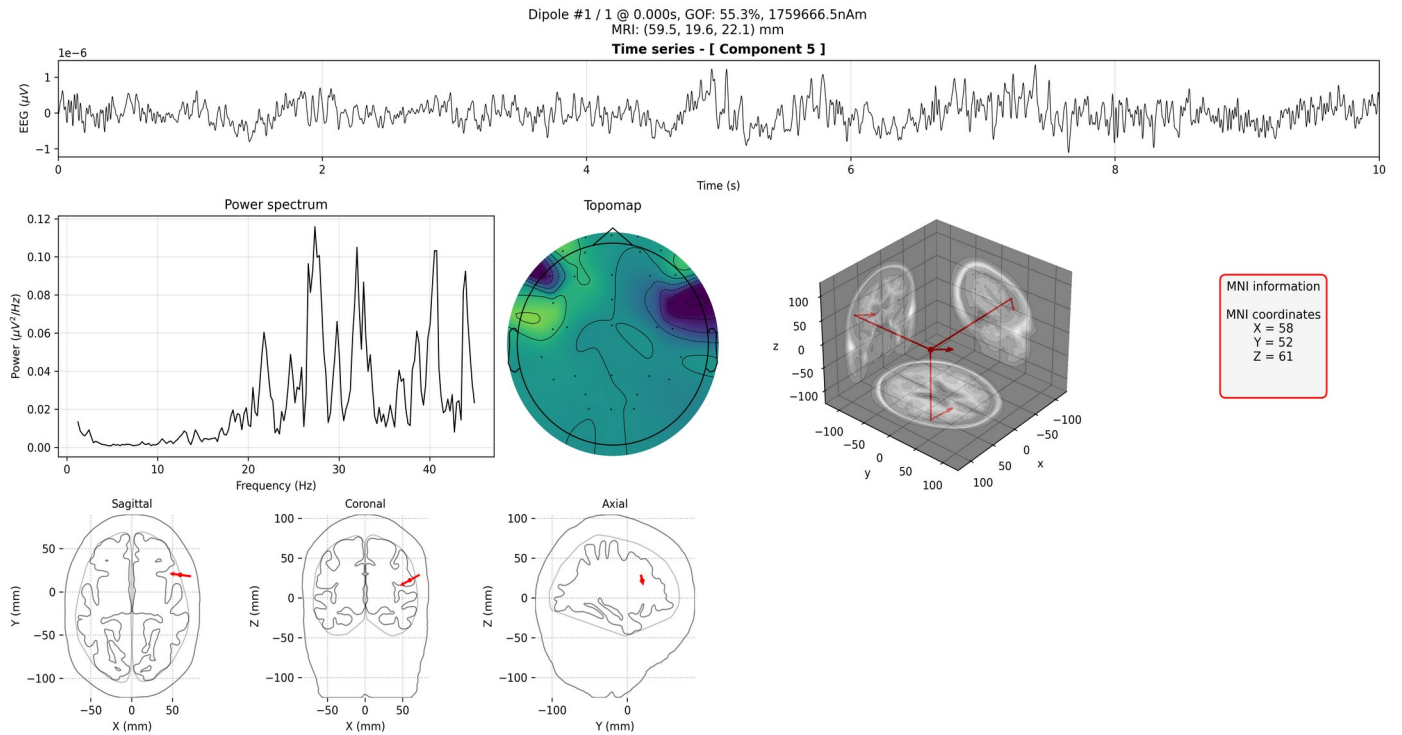


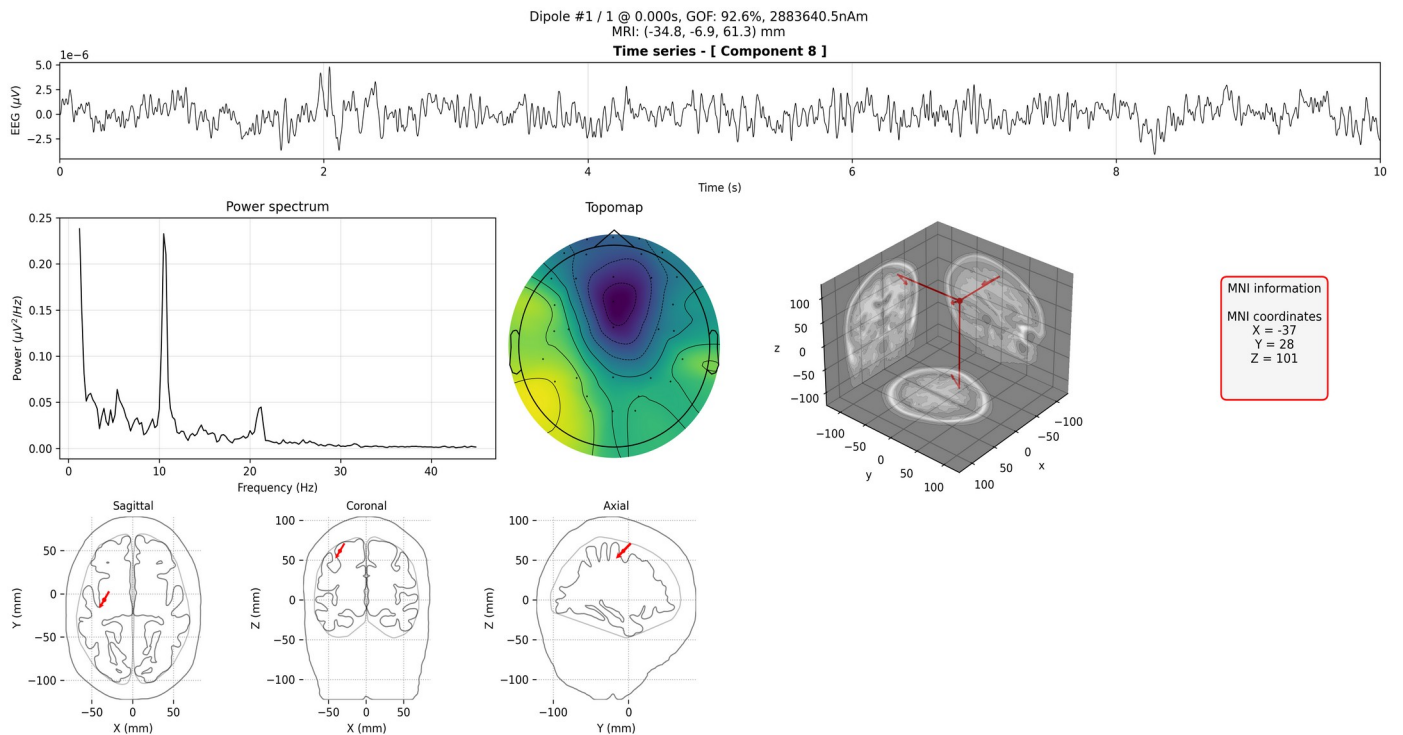
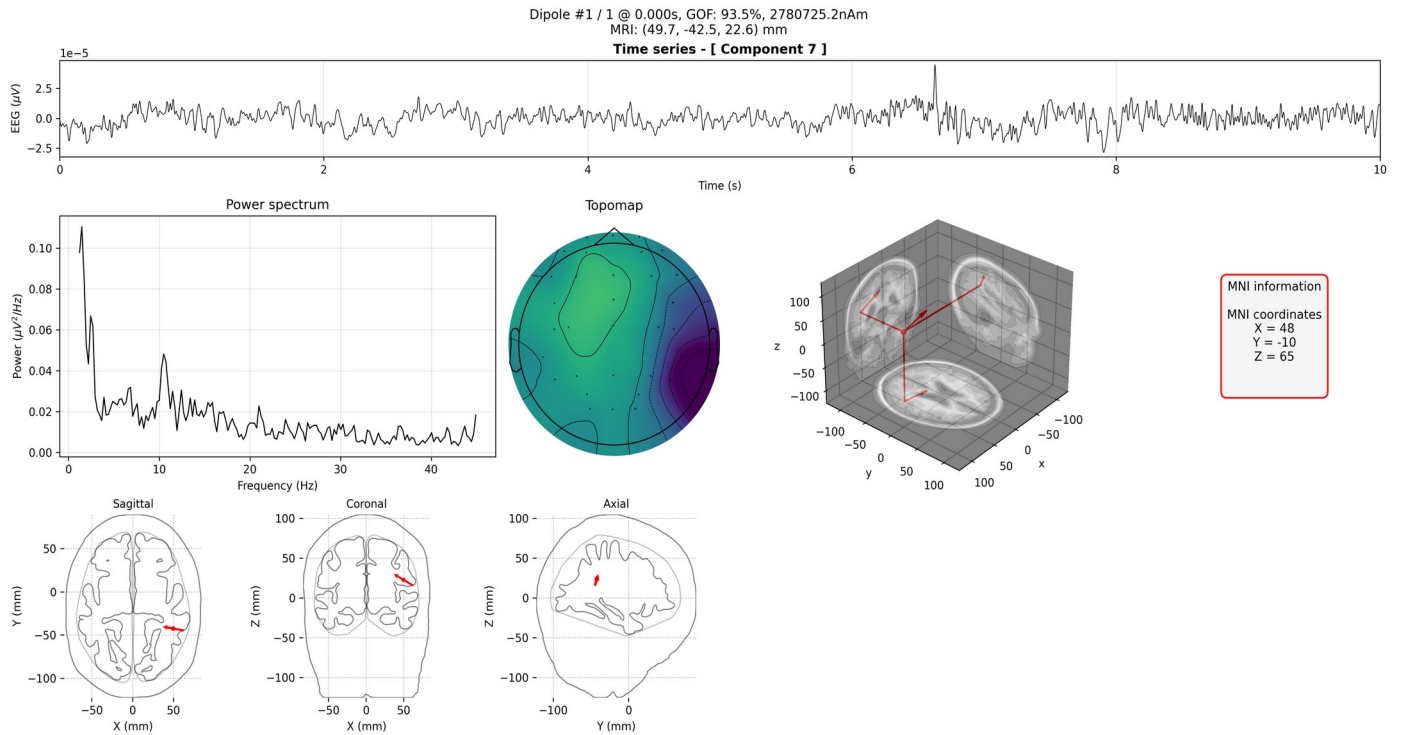
III. ICA Component Analysis

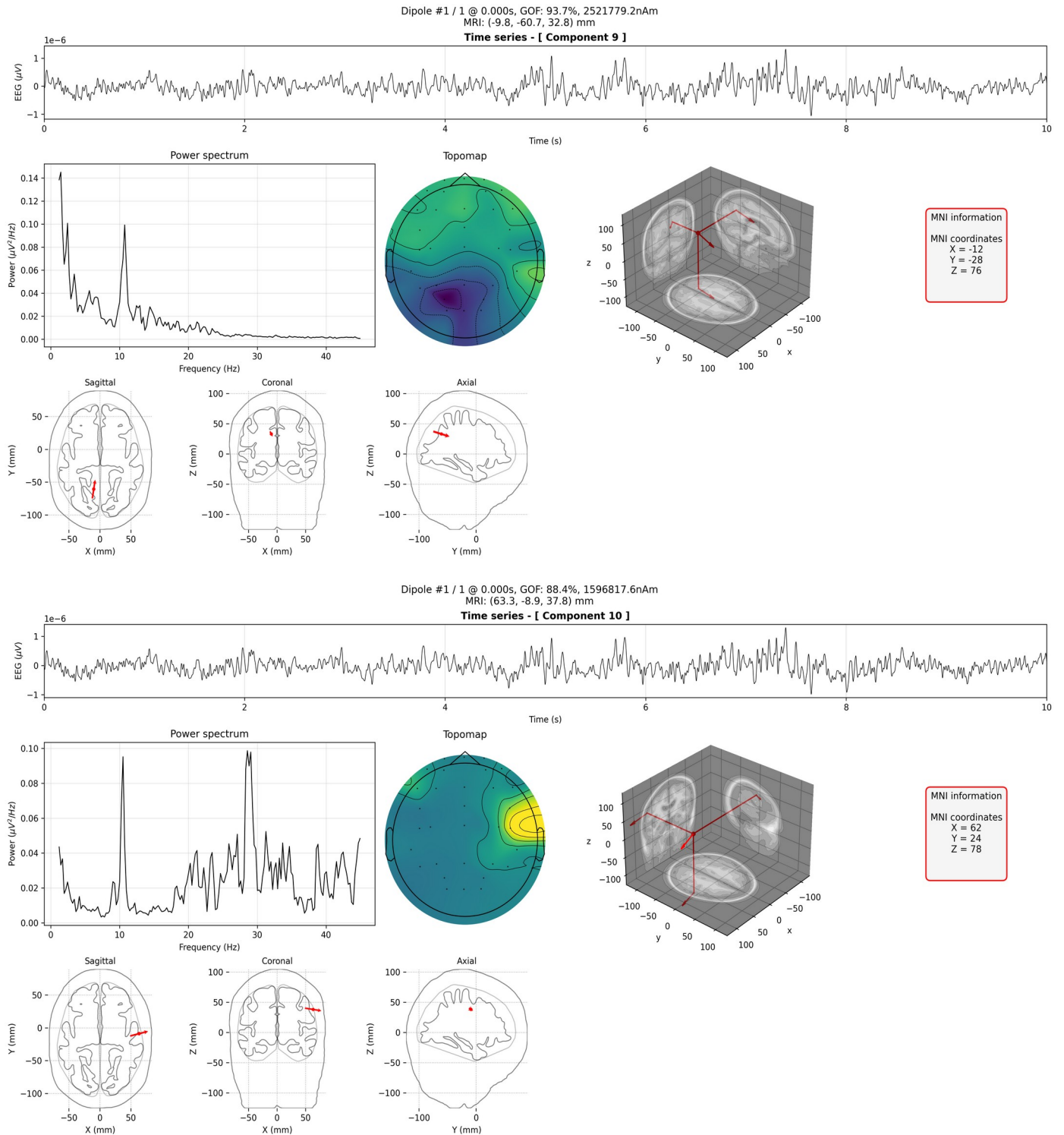
Independent component analysis (ICA) is a statistical method to separate independent sources from superimposed signals. It is the most common method that has been used in EEG data decomposition, and can be used to identify and remove the artifacts from raw EEG data. Features including time series, power spectrum density (PSD), component scalp map (Topomap), dipole source location (Source location) extracted from ICA are shown for each component.

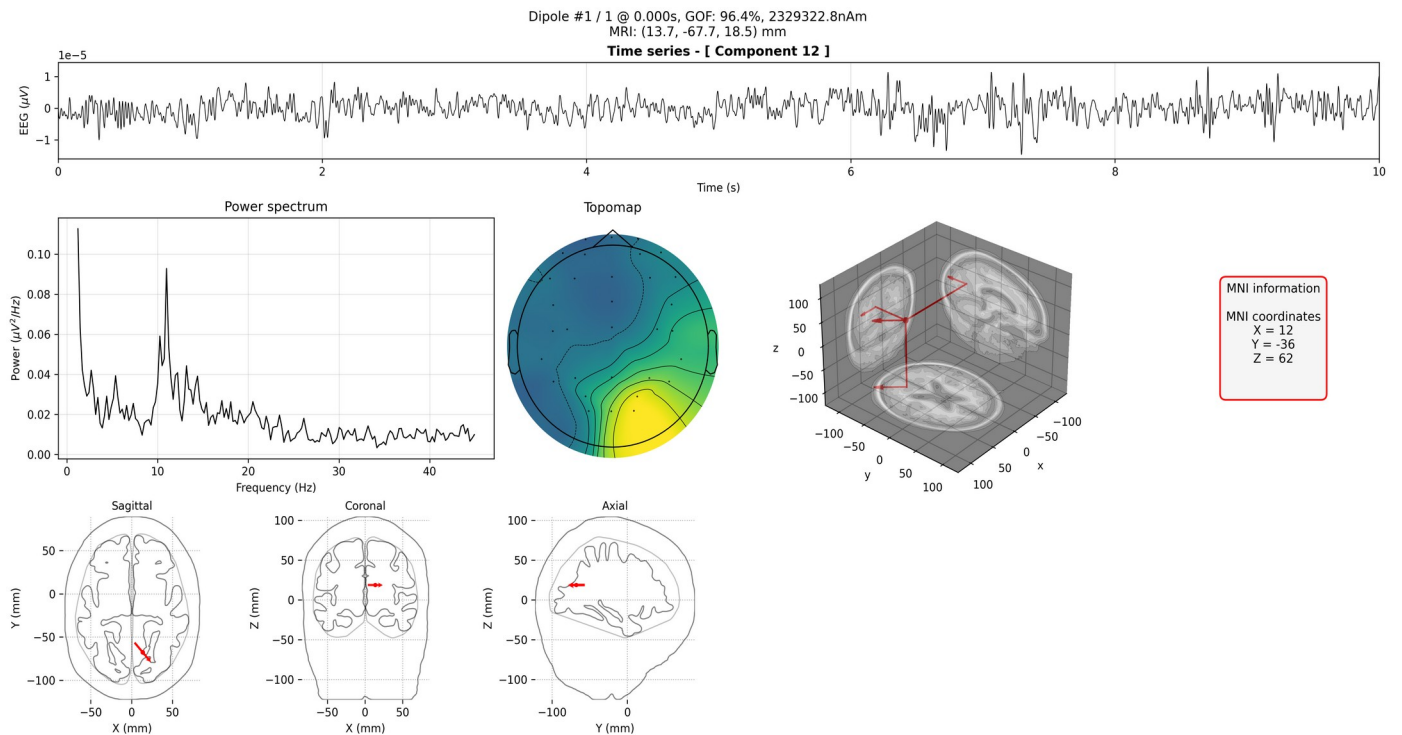
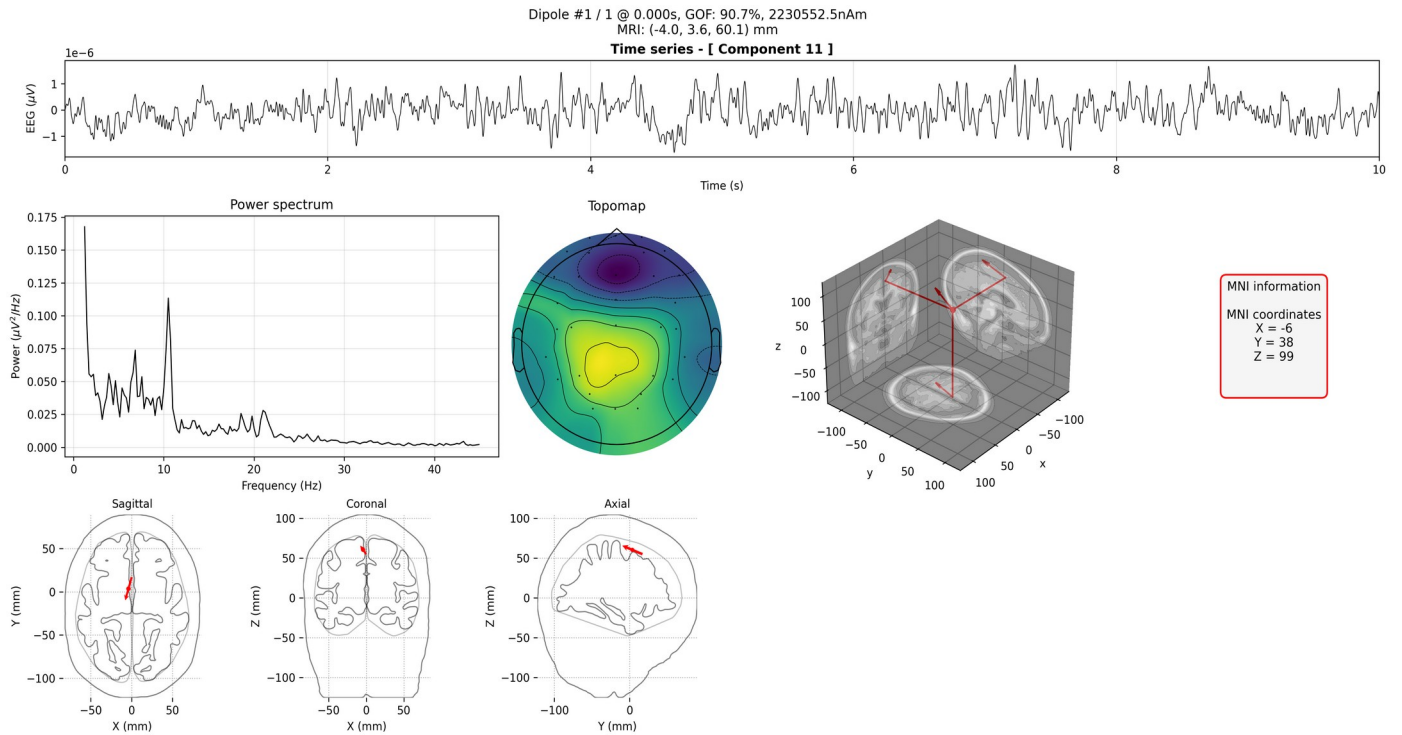


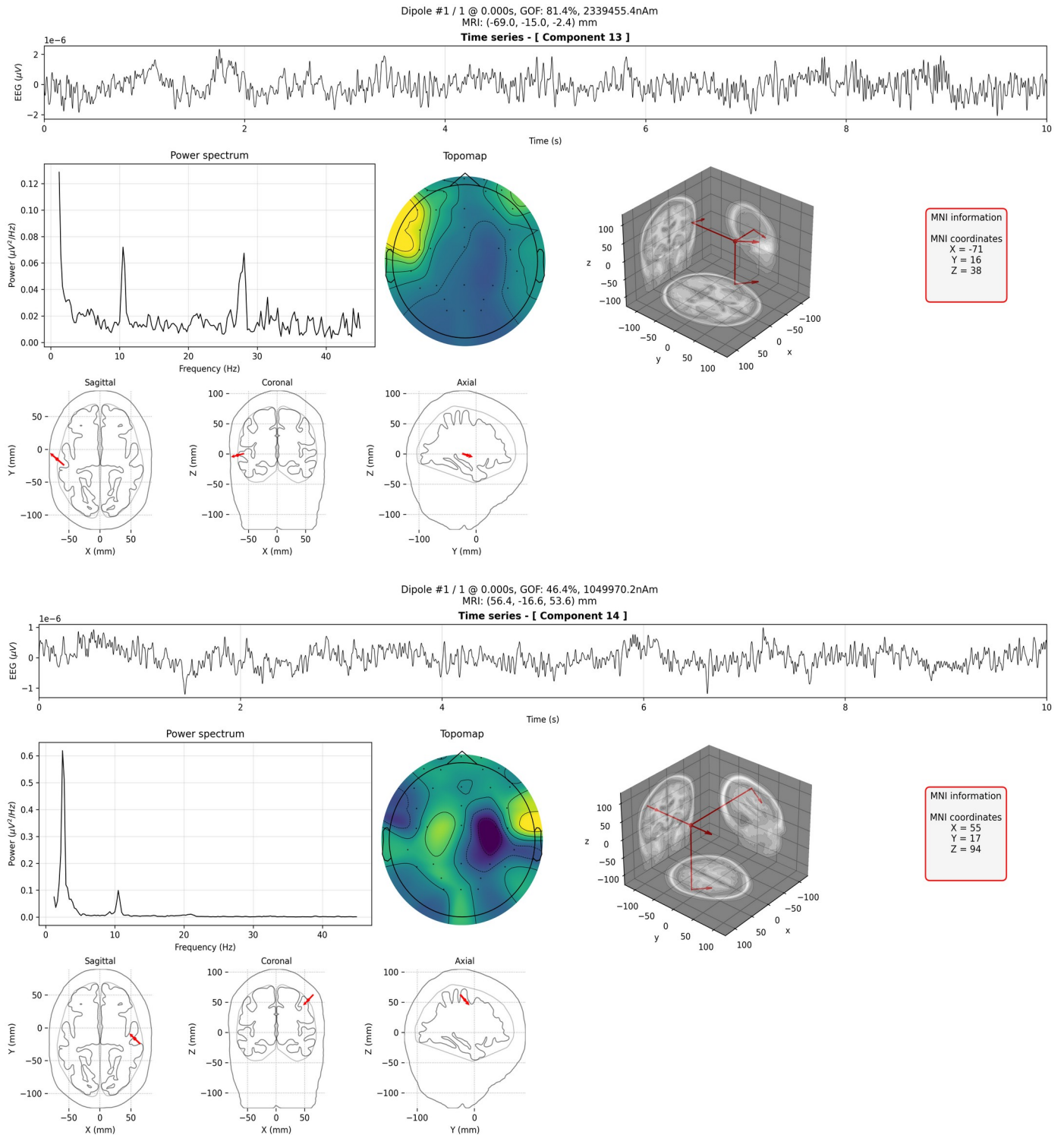


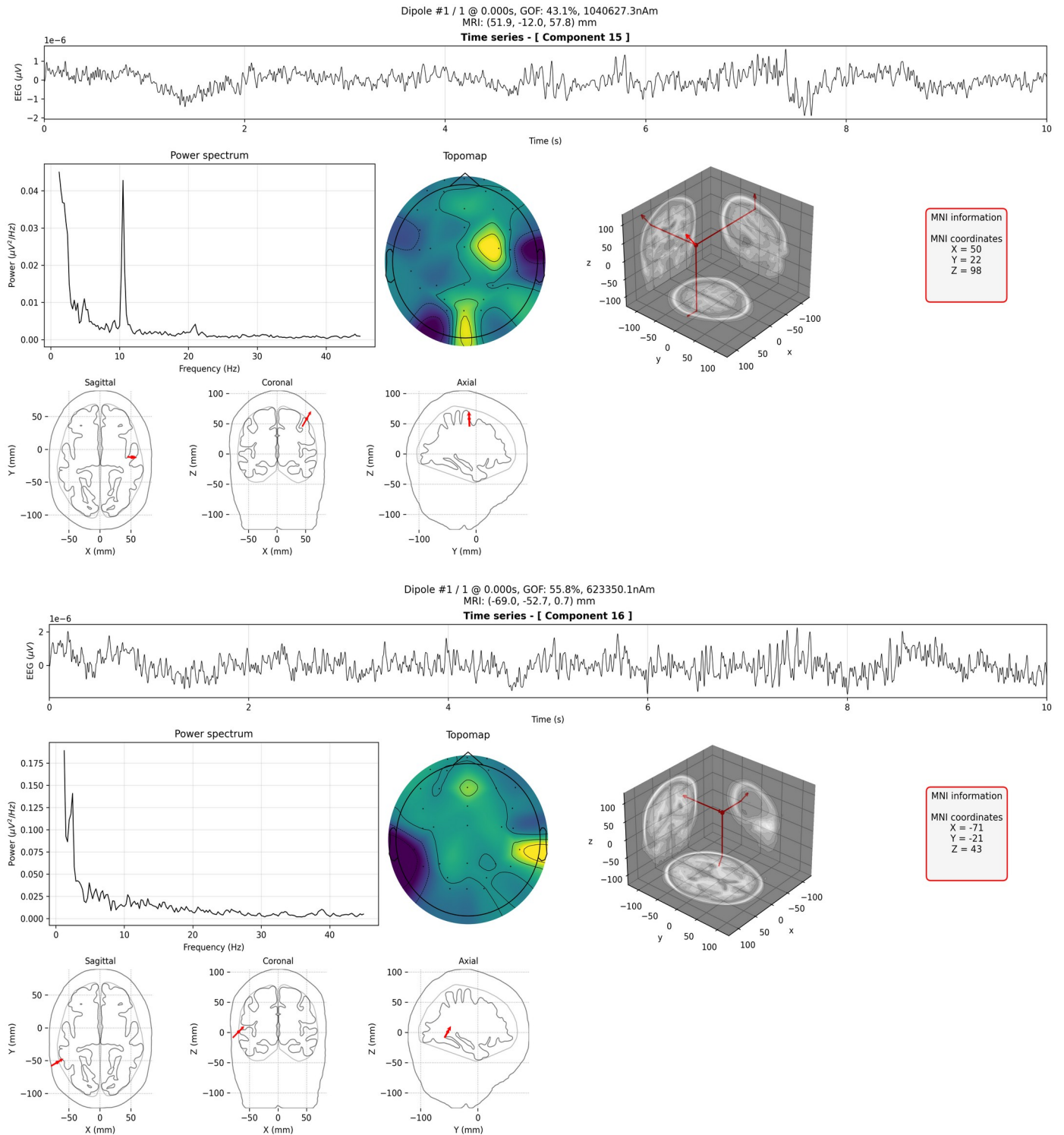


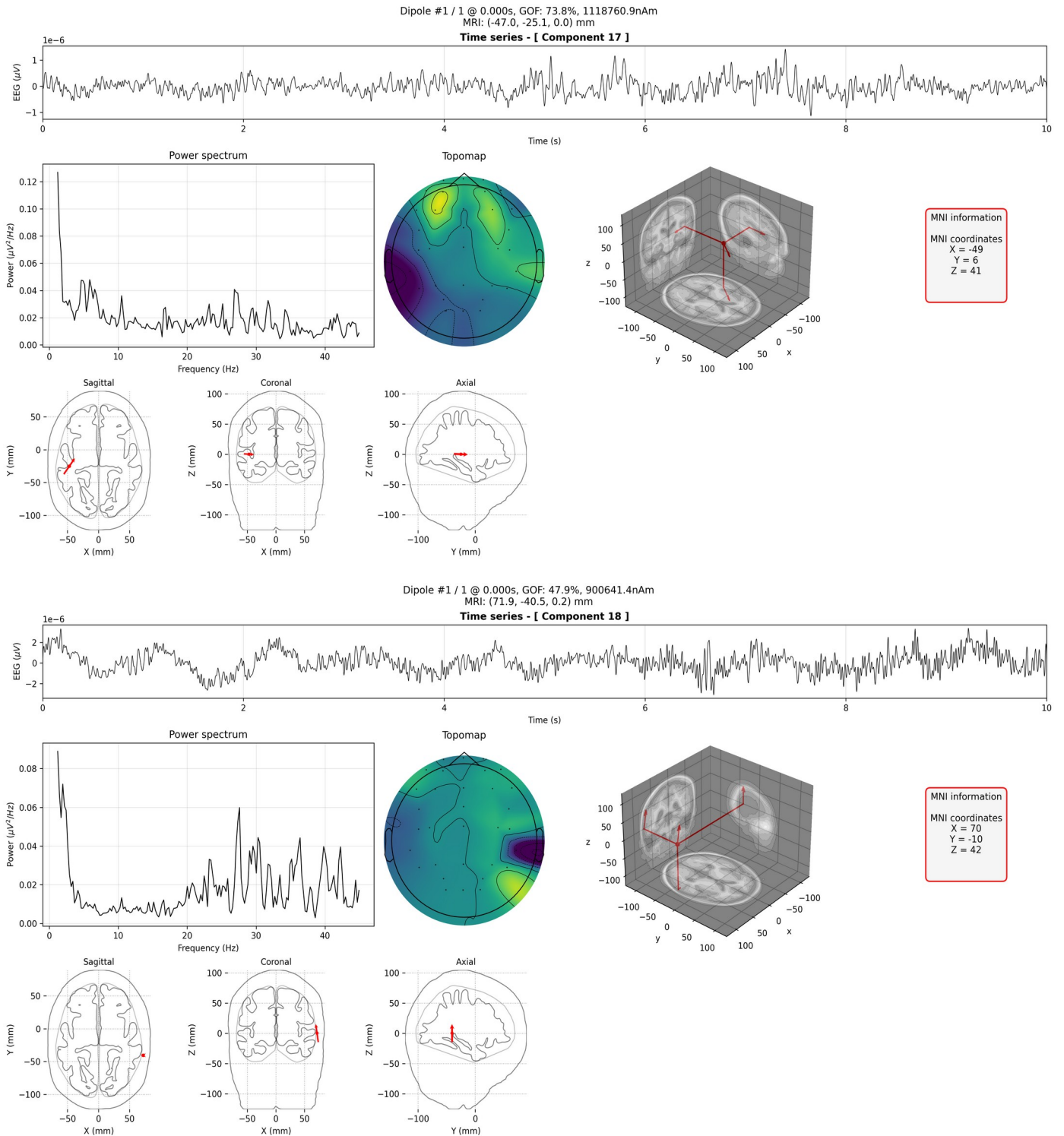


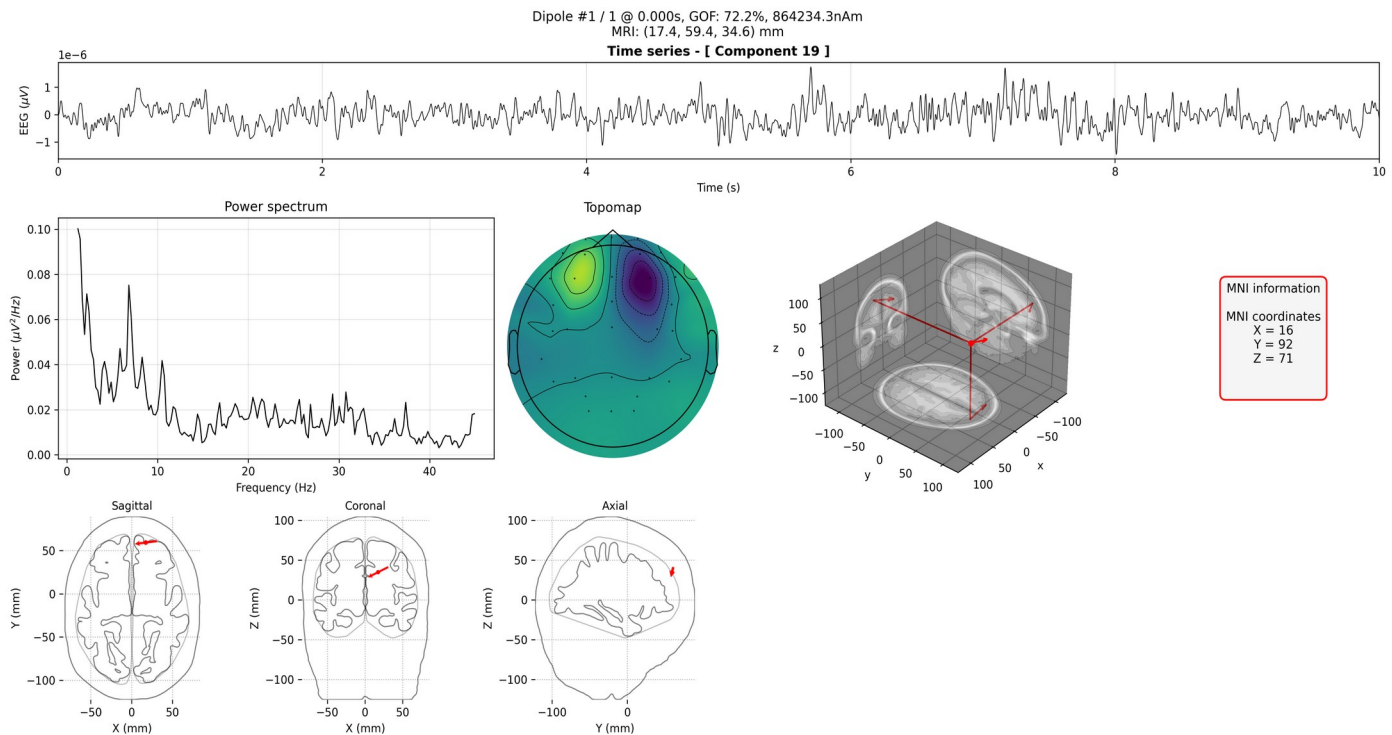










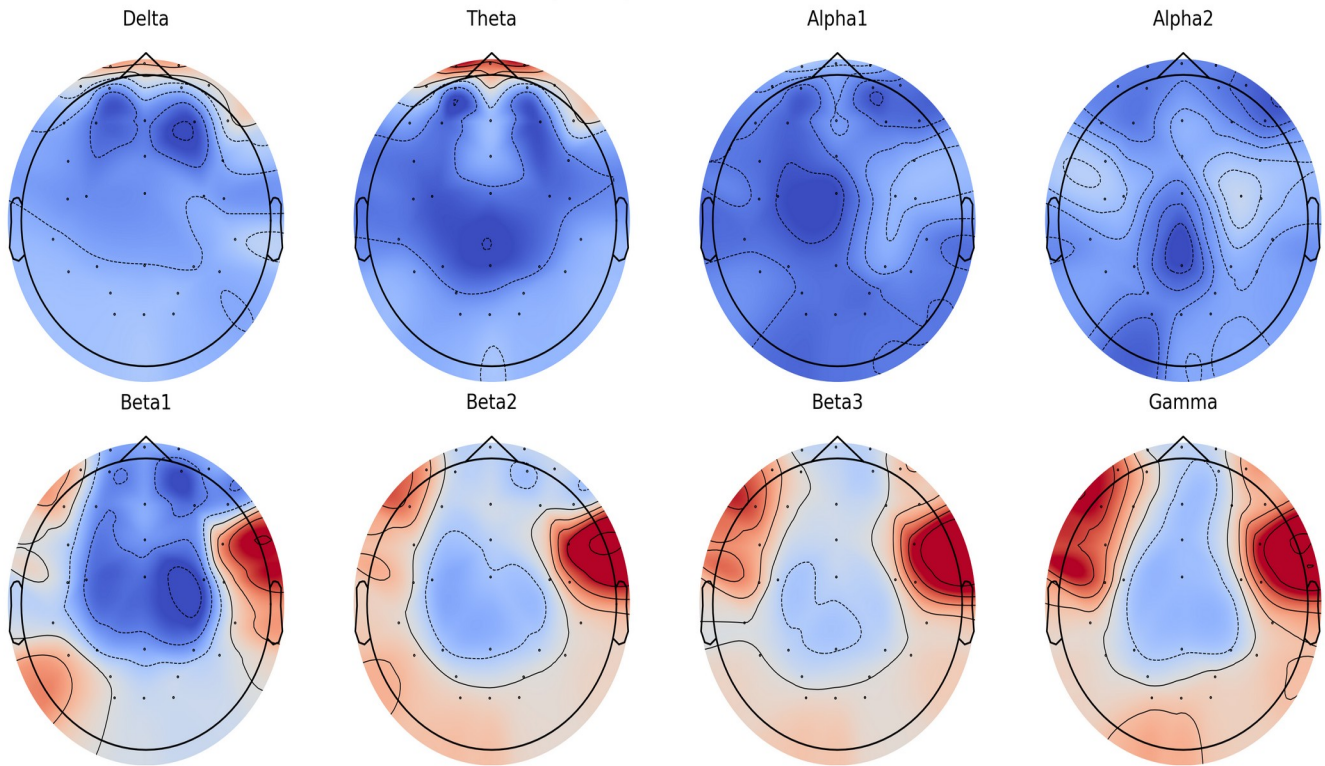


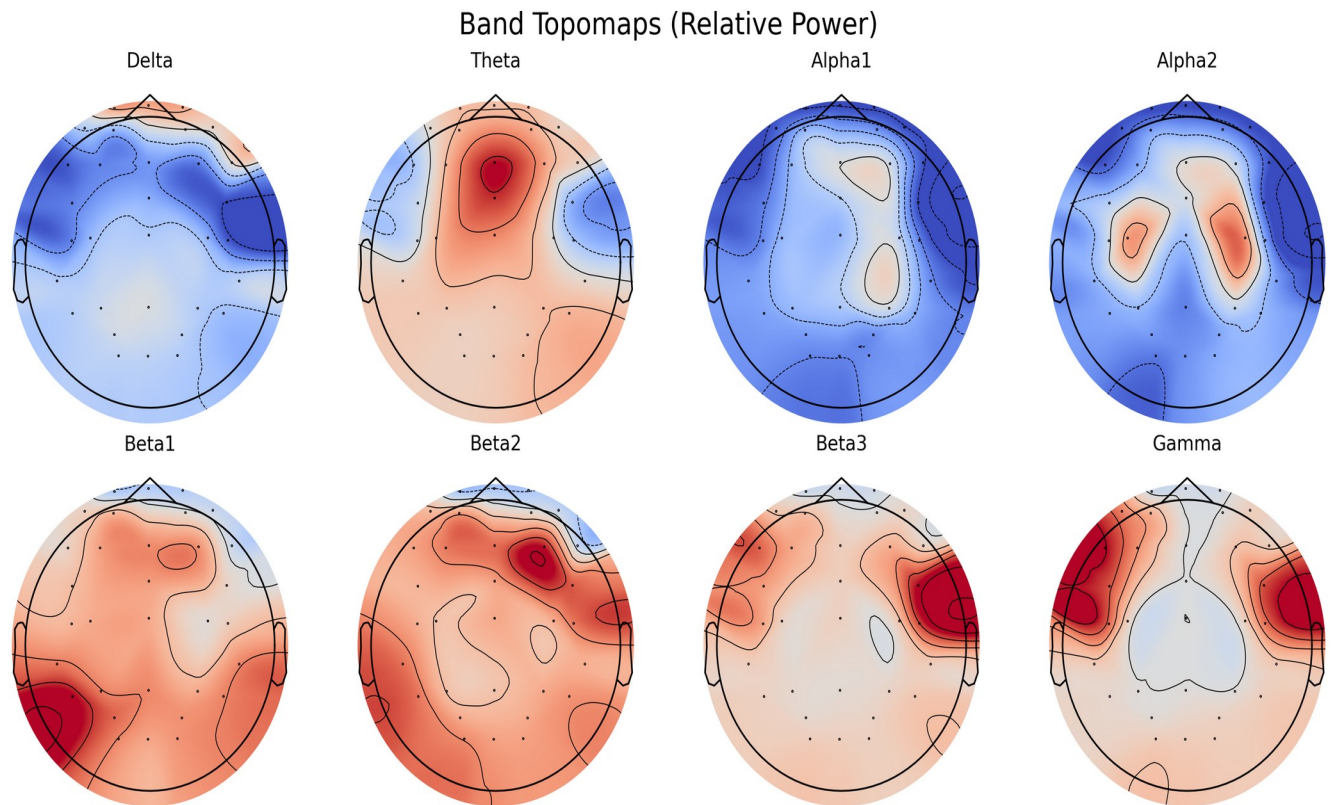
IV. Band power - Topomap

In power spectral density (PSD) 2D map, topomaps of absolute and relative power in 1 Hz bins (1 - 45 Hz) as well as each frequency band are presented. Absolute power is the sum of the component powers for each frequency band. Relative power is the absolute power in a specific frequency band divided by the total power. It is advisable to compare relative power with absolute power, since absolute power reflects the individual differences due to variations in brain tissue. This feature provides absolute and relative power based on six brain regions (prefrontal, frontal, left temporal, right temporal, central, parietal, and occipital). The power spectra for each of the 19 channels are shown in the following feature, PSD spectrum (below).

IV-I Band

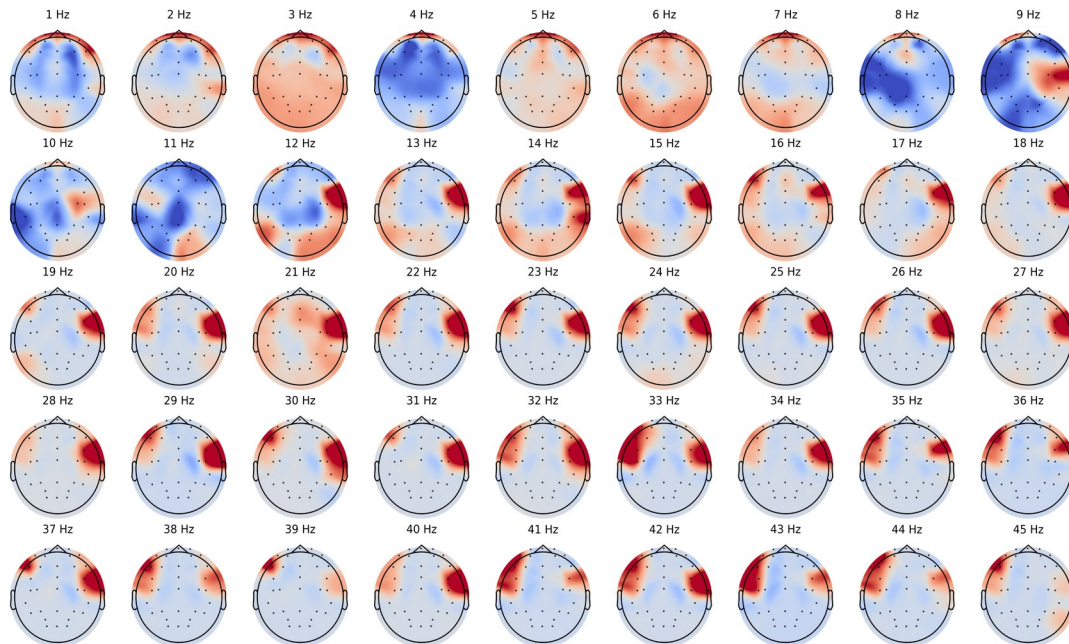
Band Topomaps (Absolute Power)





IV-II Absolute band powers (1 – 45 Hz)

Absolute Power (1 - 45hz)



IV-III Relative band powers (1 – 45 Hz)

Relative Power (1 - 45hz)

